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# Matching the wavefield is not matching the CPML solver

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**Harsh Pratap Singh**

Department of Earth Sciences  
Indian Institute of Technology Kanpur  
harshpratapsingh8210@gmail.com

**Dibakar Ghosal**

Department of Earth Sciences  
Indian Institute of Technology Kanpur  
dghosal@iitk.ac.in

**Subhojit Roy**

Department of Computer Science and Engineering  
Indian Institute of Technology Kanpur  
subhojit@iitk.ac.in

## Abstract

Learned wave models are generally judged on pressure snapshots and receiver data, and a close match is often taken as evidence that the network learned the simulator. We show that this does not hold for acoustic wave simulation with convolutional perfectly matched layers (CPML), where pressure is only part of the solver state and the absorbing layer stores memory variables that affect later updates. A surrogate can match the observed pressure field while still failing the hidden state, continuation, or derivative needed to replace that solver. We show this by comparing against the declared CPML solver at five levels, wave kinematics, receiver data, pressure and memory state, continuation, and the full waveform inversion (FWI) derivative. The clearest failure is a trained Fourier neural operator (FNO) with snapshot error around 0.13% and receiver error around 0.10%, yet completing its missing memory leaves a recurrence residual around 0.03, above both a loose  $10^{-2}$  diagnostic and the  $10^{-4}$  solver threshold. The completed state does not satisfy the declared recurrence even though the pressure evidence looks strong. The same gap reaches FWI because inversion uses the solver derivative, not only the matched pressure field. On 100 local FWI targets whose receivers sit in the CPML layer, the FNO candidate gradient, using that arm's forward map and Jacobian together, increases the exact CPML objective on 11 of them. The same test is harmful on 56 targets for a receiver-only stencil with no memory, and on 0 for exact CPML and for a full-state readout with the correct tangent. So pressure agreement is not enough evidence to replace the solver, restart from it, or take an FWI step with its derivative.

## 1 Introduction

Learned forward simulators are usually judged by visible outputs such as pressure snapshots, receiver data, and error tables. A close match can make a surrogate look like the solver it was trained to copy, but a solver is also used to restart from an intermediate time and to provide derivatives for inversion. Those uses depend on the state carried between time steps, not only on the pressure field that is easy to plot.

In this work we study that gap on acoustic wave simulation with convolutional perfectly matched layers (CPML). This absorbing layer damps outgoing waves so a small grid can look unbounded, and it stores extra memory variables at the edge to do that. Perfectly matched layers were introduced

for electromagnetics [Bérenger, 1994]. The convolutional memory form is CPML [Roden and Gedney, 2000], and the seismic unsplit pressure formulation is due to Komatitsch and Martin [2007]. The complete discrete state contains current pressure, previous pressure, and those auxiliary boundary-memory variables. The memories affect later pressure updates near the boundary. Thus, a learned pressure predictor can match pressure plots and receiver data while still missing the state needed to continue the declared CPML solver.

The same state issue matters for inversion, because full waveform inversion (FWI) fits a subsurface model so that simulated receiver data match the recorded data [Tarantola, 1984, Virieux and Operto, 2009]. The forward solver and its derivative set the data misfit, the objective, and the update direction used by the optimiser [Plessix, 2006]. If a learned map matches receiver data but has the wrong CPML state or derivative, the optimiser is no longer using the declared CPML map. In our test, the candidate gradient, which uses both the candidate forward map and its Jacobian, can make the exact CPML objective go up instead of down.

What we found is easiest to see on one trained Fourier neural operator (FNO). It looks accurate on snapshots and receivers, but after we fill in the missing CPML memory the complete recurrence no longer holds. When we use that candidate gradient for one local FWI step on 100 targets whose receivers sit in the absorbing layer, the exact CPML objective rises on 11/100 of them. A receiver-only stencil with no memory does worse on the same targets (56/100), while exact CPML and a full-state readout with the correct tangent stay at 0/100. The exact-CPML zero follows from the shared step rule. Those numbers compare checks and uses on this CPML setup, not neural wave architectures against each other.

We turn this gap into a verification profile for a fixed acoustic CPML solver, scoring whether the evidence supports restart or differentiation rather than treating a pressure match as a reason to substitute the surrogate for the solver. The rest of the paper defines that profile, explains why the hidden CPML state and the FWI tangent matter, and then tests what happens when weaker validation is promoted to an FWI decision. Settings are in Appendix B.

## 2 Related work

A lot of papers bound PINN and neural PDE solver error from residual information, some at sampled points and some over a continuous domain [Liu et al., 2023, Qian et al., 2023, Mishra and Molinaro, 2023, Mukherjee et al., 2026, Eiras et al., 2024]. Later papers try to verify that the network is solving the PDE, not only bound how large the residual is. Ellinas et al. [2024] verify networks that approximate differential equations by bounding residual quantities and propagating those bounds. Haugen et al. [2025] give a posteriori checks for neural PDE solvers from network norms, derivative norms, and residual bounds. PINNs can still fail in training even when the loss is a residual [Raissi et al., 2019, Krishnapriyan et al., 2021]. Those papers ask whether the residual is small, or whether the predicted field is close to the true solution. We ask whether the learned map should be used to restart the run or to take a derivative. For a CPML solver claim, the residual that answers that is the discrete recurrence, including the absorber-memory rows. You can also project or correct a predicted field so the residual gets smaller [Rochman Sharabi and Louppe, 2025, Huang and Perdikaris, 2026]. The corrected field may then satisfy the CPML map, but that does not show that the original learned pressure model implements CPML.

The same issue appears for derivatives, because FWI uses the solver derivative, not only a small residual. Cao et al. [2026] verify explicit derivative kernels and discretised PDE adjoints numerically, then embed those kernels in an inversion workflow. That is implementation verification of a derivative, not a check that a learned map matches a declared CPML tangent. For a learned CPML surrogate and a declared finite-difference CPML recurrence, we ask whether the candidate supplies the hidden absorber state, continuation from that state, and the wavespeed tangent of that recurrence.

The wave papers usually score the plotted field instead. FNO and DeepONet [Li et al., 2021, Kovachki et al., 2023, Lu et al., 2021] are the operator architectures used here. FNO-style models are used as wave surrogates [Liu et al., 2024, Yang et al., 2021], including frequency-domain Helmholtz FNO [Li et al., 2023] and frequency-domain acoustic VTI FNO [Konuk and Shragge, 2021], and as FWI forward maps [Yang et al., 2021, Zou et al., 2025]. Those papers treat pressure and receiver accuracy as the main evidence, and we keep those as valid observation-level checks. Some of them also backpropagate through the learned map for inversion, but they do not test the hidden absorber

state, continuation under a declared CPML recurrence, or whether that learned tangent matches the CPML wavespeed derivative. OpenFWI is a public inversion-dataset benchmark with convolutional inversion baselines, not an FNO or DeepONet solver study [Deng et al., 2022]. Changing the boundary condition can also change what a learned PDE solver represents [Shikhman, 2026], and the present case is stricter still because even with one fixed CPML recurrence a pressure-only surrogate can miss the complete state needed for restart and differentiation.

### 3 Verification profiles for earned claims

A verification profile reports the strongest scientific use supported by the evidence for a fixed acoustic CPML solver. In this setup we fix the declared solver, the receiver measurements, how the complete state is packed, and the acceptance tolerances, then score wave kinematics, receiver data, pressure and memory state, continuation from that state, and the FWI derivative. For a candidate  $f$ , we write

$$V(f) = (v_{\text{kin}}, v_{\text{obs}}, v_{\text{state}}, v_{\text{cont}}, v_{\text{jac}}),$$

where  $v_{\text{kin}}$  checks wave kinematics,  $v_{\text{obs}}$  checks receiver data,  $v_{\text{state}}$  checks pressure and memory state,  $v_{\text{cont}}$  checks continuation from that state, and  $v_{\text{jac}}$  checks the derivative used by FWI. Each component is PASS, FAIL, unmeasured (written U in Table 1), or MIXED. A pass means the submitted fields satisfy the declared check, a fail means they violate it, and unmeasured means the needed field was not supplied, which is different from passing a weaker test. When held-out rows disagree we label a component mixed. The reported use is the strongest use whose required earlier checks all pass.

The declared physics is the variable-density acoustic wave equation

$$\partial_{tt}u - c^2(\mathbf{x})[\Delta u - \nabla(\log \rho) \cdot \nabla u] = s(\mathbf{x}, t), \quad (1)$$

with zero initial conditions and a declared absorbing-boundary realisation. The  $\nabla(\log \rho)$  term is the usual variable-density correction written compactly as  $\nabla \rho / \rho$ . The reference solver is the Fortran SEISMIC\_CPML pressure formulation [Komatitsch and Martin, 2007, 2018]. Its complete discrete state contains consecutive pressure levels and CPML memory variables. We write  $\mathcal{Z}_h^{\text{CPML}}$  for that complete solver, taking  $(c, \rho, s)$  to the trajectory  $z_h$ , and  $\mathcal{G}_h^{\text{CPML}} = \Pi_p \mathcal{Z}_h^{\text{CPML}}$  for the pressure field that receivers see.

Table 1 summarises the main experiment rows in this vocabulary. Section 4 explains the scientific failures behind those rows.

Table 1: The table organises the main experiment rows by the strongest use the evidence supports. The profile order is kinematics, observation, complete state, continuation, and FWI derivative. P means pass, F means fail, U means the needed field was not supplied, and M means mixed held-out results. The FWI harm counts are separate downstream tests on whether one matched step raises the exact objective. They are not additional profile entries. Exact CPML’s zero count follows from the shared step rule. Wilson intervals are nominal binomial intervals on this selected sample, not a survey rate.

| Candidate                         | Profile   | Strongest earned claim | Downstream action test                                                  |
|-----------------------------------|-----------|------------------------|-------------------------------------------------------------------------|
| Exact CPML                        | P/P/P/P/P | FWI derivative         | 0/100 harmful steps, by the step rule                                   |
| Full-order readout                | P/P/P/P/P | FWI derivative         | 0/100 harmful steps, Wilson 95% upper 3.7%                              |
| FNO pressure surrogate            | P/P/F/U/F | Observation only       | residual $2.83 \times 10^{-2}$ , 11/100 harmful                         |
| Residual-trained FNO              | P/P/F/U/U | Observation only       | residual $6.61 \times 10^{-3}$ to $6.72 \times 10^{-3}$ , FWI not rerun |
| State-aware recurrent operator    | P/M/F/U/F | Kinematics only        | residual 8.81 to 9.31, tangent 0.160 to 0.346                           |
| Memory-free receiver-only stencil | P/P/F/U/F | Observation only       | state and tangent fail, 56/100 harmful                                  |

**Lemma 1** (CPML memories are the continuation state). *Fix the grid, material, CPML coefficients, and future source schedule. If  $\eta^n$  denotes the stacked CPML auxiliary memories, then  $x^n = (u^n, u^{n-1}, \eta^n)$  is a Markov continuation state for the declared CPML recurrence. A pressure-only pair is not guaranteed to be Markov, because hidden absorber memories can feed back into the next pressure update. A complete pressure archive can replay the memories from the initial state, but that archive is not a finite continuation state.*

As CPML memories store earlier boundary information, two states can have the same recent pressure fields but different absorber memories, and those memories can change the next pressure

update. If the memory has mostly decayed, a pressure-only history may behave approximately like a continuation state, but that is an empirical fact to check, not something receiver agreement proves. Appendix A gives the recurrence expansion and the  $q$ -window replay.

We use four relative errors to keep visible agreement separate from solver agreement. These are declared scoring quantities, not new PDE identities. The reference object is the CPML solution  $z_h$ , and the candidate object is  $z_\theta$ . The operator  $\Pi_p$  extracts pressure from a complete state,  $P_R$  extracts the receiver data,  $S_Z$  sets how complete-state components are weighted, and  $D_m$  means differentiation with respect to the physical model parameters. Receiver mismatch  $E_R$  compares receiver data, complete-state mismatch  $E_Z$  compares the full pressure and memory trajectory, receiver-data derivative mismatch  $E_{J,R}$  compares receiver-data tangents in a model direction  $v$ , and complete-state derivative mismatch  $E_{J,Z}$  compares complete-state tangents in the same direction.

For a candidate complete trajectory  $z_\theta$ , receiver operator  $P_R$ , and model direction  $v$ , we report

$$\begin{aligned} E_Z &= \frac{\|S_Z(z_\theta - z_h)\|_2}{\|S_Z z_h\|_2 + \epsilon}, & E_R &= \frac{\|P_R(\Pi_p z_\theta - \Pi_p z_h)\|_2}{\|P_R \Pi_p z_h\|_2 + \epsilon}, \\ E_{J,R}(v) &= \frac{\|P_R(D_m \Pi_p z_\theta - D_m \Pi_p z_h)[v]\|_2}{\|P_R D_m \Pi_p z_h[v]\|_2 + \epsilon}, & E_{J,Z}(v) &= \frac{\|S_Z(D_m z_\theta - D_m z_h)[v]\|_2}{\|S_Z D_m z_h[v]\|_2 + \epsilon}. \end{aligned} \quad (2)$$

$E_R$  is the visible receiver check,  $E_Z$  is the complete pressure-and-memory check, and the two Jacobian errors do the same split for the derivative. The 10% cut is a reporting threshold for this packed geometry. It helps the table separate visible agreement from complete-state agreement, and it is not a field-scale physical tolerance. The normalisations in Eq. (2) are fixed before scoring, and  $\epsilon > 0$  prevents division by a nearly zero denominator. In all reported  $31 \times 31$  percentage tables,  $S_Z = I$ , so pressure and CPML memory variables are compared in the packed recurrence coordinates. A different packing would define a different  $S_Z$  and a different link between residual size and complete-state error. Solver use also requires the row-scaled recurrence residual and the tangent checks described below.

The residual check asks whether the declared CPML time stepper could have generated the entire pressure and memory trajectory. We hold the grid, material model, source schedule, and CPML coefficients fixed, and pack the pressure levels and CPML memories over the scored horizon into  $z = (p^0, \dots, p^N, \eta^0, \dots, \eta^N)$ . Each solver update becomes one row of a linear system on that stacked trajectory. The initial-condition rows pin the starting state, the memory rows enforce the CPML recurrences, the pressure rows enforce the finite-difference wave update that uses those memories, and the boundary rows enforce the declared boundary values. The declared solver is

$$A_h(m)z = b_h(m, s), \quad (3)$$

where  $A_h$  is the CPML recurrence, including the absorber-memory rows,  $z$  is the complete discrete state, and there is no learned fit or different PDE. Before scoring, we set  $D$  to the inverse  $\ell_2$  norm of each assembled recurrence row so that source, boundary, pressure-update, and memory-update equations receive comparable numerical weight. A pressure-only residual leaves those memory rows out, so it cannot certify continuation under Eq. (3). For this representation,  $10^{-4}$  is the solver threshold used throughout the paper, while  $10^{-2}$  remains a loose diagnostic level. On the retained  $31 \times 31$ , 18-update homogeneous calibration, a numerical singular-value estimate in Eq. (4) maps the  $10^{-4}$  residual threshold to about 1.3% complete-state error. That estimate is not a certified lower bound, and that operator is not the 27-update heterogeneous FWI gate. Details are in Appendix Table 2.

Agreement on arrival times supports kinematics only, identically processed receivers support receiver-data use, and complete-state closure supports the corresponding forward map. Receiver and state-tangent closure on declared model directions is required before treating the learned map as an FWI derivative. Exact receiver data do not bound state error, and exact state at one model does not bound the Jacobian (Appendix A, Lemma 2).

Equation (4) is the standard residual bound for Eq. (3) after row scaling. Since the reference trajectory satisfies that system, invertibility of the initialised recurrence gives a complete-state error certificate when  $\sigma_{\min}(DA_h S_Z^{-1})$  is known.

$$\|S_Z(z_\theta - z_h)\|_2 \leq \sigma_{\min}(DA_h S_Z^{-1})^{-1} \|D(A_h z_\theta - b_h)\|_2. \quad (4)$$

This bounds complete-state error by the scaled residual of Eq. (3). A pressure residual that omits memory equations does not control Eq. (4), because it is not the residual of  $A_h$ . The reported

experiments still score residual,  $E_Z$ ,  $E_R$ , and  $E_{J,Z}$  directly, because the state residual certificate does not replace tangent or downstream FWI checks.

For inversion we need that residual as a derivative, not only as a state check. Differentiating the declared recurrence in a model direction produces a tangent residual, and the same invertibility argument bounds the Jacobian error.

**Corollary 1** (Fixed-discrete tangent check). *Let  $A_h(m)z_h(m) = b_h(m)$  be differentiable and nonsingular near  $m$ , and fix a direction  $v$ . Write  $\dot{A}_h = D_m A_h(m)[v]$ ,  $\dot{b}_h = D_m b_h(m)[v]$ , and  $\dot{z} = D_m z(m)[v]$ . For an approximate pair  $(z_\theta, \dot{z}_\theta)$ , the differentiated residual  $r_{J,h} = A_h \dot{z}_\theta + \dot{A}_h z_\theta - \dot{b}_h$  satisfies*

$$\|S_Z(\dot{z}_\theta - \dot{z}_h)\|_2 \leq \sigma_{\min}(DA_h S_Z^{-1})^{-1}(\|Dr_{J,h}\|_2 + \|D\dot{A}_h S_Z^{-1}\|_2 \|S_Z(z_\theta - z_h)\|_2). \quad (5)$$

*State closure together with a vanishing differentiated-recurrence residual is sufficient for directional tangent closure at the fixed discretisation, which is why the FWI derivative needs its own check and cannot be inferred from pressure agreement alone.*

Vanishing complete residual does not control  $r_{J,h}$ . For inversion, write  $\Phi$  for the exact CPML receiver objective and  $\tilde{\Phi}$  for the candidate objective. Their gradients differ by

$$\nabla \tilde{\Phi} - \nabla \Phi = (\tilde{J} - J)^T W(F - d) + \tilde{J}^T W(\tilde{F} - F). \quad (6)$$

where  $J$  and  $\tilde{J}$  are the Jacobians of the exact and candidate receiver maps. Both the forward mismatch and the Jacobian mismatch can increase the exact objective. The retained FWI test uses each arm's own forward values together with its own Jacobian, so a harmful step is not identified as derivative error alone. Related inexact constrained-optimisation work studies how approximate solves and derivative information affect optimiser convergence [Heinkenschloss and Vicente, 2002].

## 4 Experiments

The experiments follow the order in which the failure appeared. First we ask whether ordinary validation would have made the trained pressure surrogate look usable as a solver. Then we ask whether that extra interpretation changes an FWI step, and what continuation looks like when the complete absorber state is carried. Settings are in Appendix B.

### 4.1 Can ordinary validation overclaim a learned solver?

The gap appears when a pressure predictor is treated as a solver. A trained FNO2d surrogate [Li et al., 2021] predicts pressure only and has no CPML memory labels. On a held-out 24-dimensional wavespeed case, it looks accurate by ordinary pressure evidence, with 0.125% snapshot error and 0.096% receiver-data error. We then reconstruct the missing CPML memories from the predicted pressure by replaying the declared memory recurrence. The completed state still passes the 10% reporting cut, with 0.253% state error and 4.79% state-tangent error, so a 10% complete-state cut would still accept this case even though the recurrence does not close.

The complete-recurrence check changes the conclusion by rejecting the same completed state. Its residual is 0.0283, above both the loose  $10^{-2}$  diagnostic and the declared  $10^{-4}$  solver threshold. Adding complete residual to the pressure loss improves retained held-out residuals to 0.00672 and 0.00661, still short of the CPML threshold. The issue is not receiver-data error but failure of the complete recurrence, including the memory rows. Figure 1 shows this row.

A POD-DeepONet-style pressure operator [Lu et al., 2021] fails the same  $10^{-4}$  residual on all three held-out cases, with worst residual 0.0130 (Appendix C). A complete-state recurrent operator carries memory heads better than a zero-memory restart, so it is not the same failure as a pressure-only model, but its recurrence residual remains 8.81 to 9.31 and its state-tangent error remains 0.160 to 0.346. Better memory is still only as strong as the recurrence and tangent checks, and those checks have not passed.

Ordinary screens pass; causal CPML completion still fails 1e-2 residual

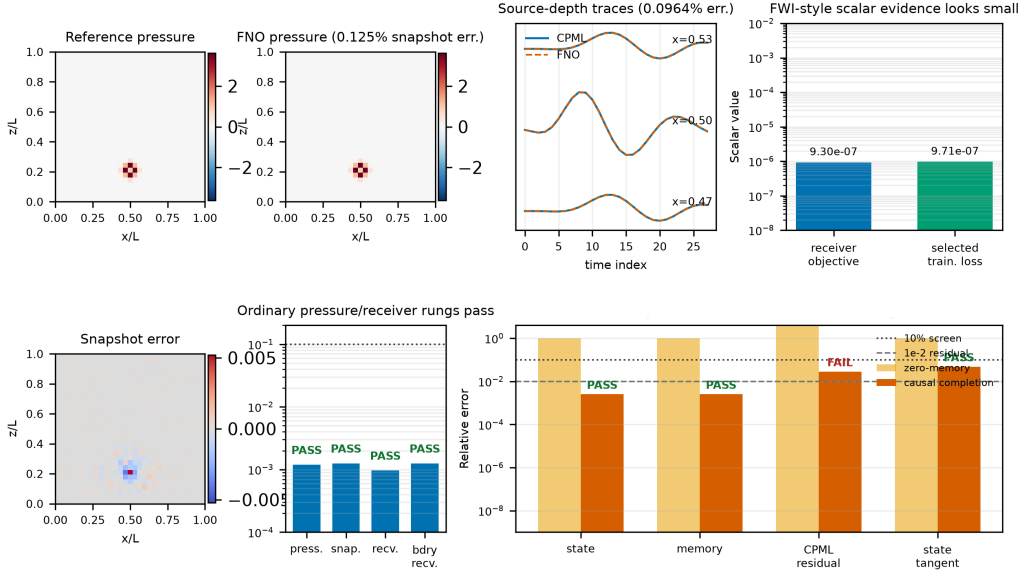


Figure 1: A trained FNO pressure surrogate passes ordinary snapshot, receiver-data, and scalar checks on a test case not used for training. After causal CPML memory completion, the complete recurrence residual is 0.0283, which fails the loose  $10^{-2}$  diagnostic and the declared  $10^{-4}$  solver threshold. State, memory, and state-tangent errors still pass the 10% reporting cut, so a 10% complete-state cut alone would not reject this row.

#### 4.2 Does overclaiming change an FWI decision?

If the FNO is treated as a solver, the next scientific use is inversion, and inversion uses the derivative of the declared map rather than the plotted pressure field. The test uses 100 local targets whose receivers sit in the CPML layer on the same 24-coefficient wavespeed family as the FNO experiment (Figure 2). We compare exact CPML, a full-state readout with the correct tangent, a receiver-only stencil with memories filled from pressure, and the FNO candidate gradient after the same memory completion. That gradient uses the FNO forward map and its Jacobian together.

Each method forms a gradient  $\tilde{g}$  on the 24-coefficient vector  $q$  and a unit descent direction  $\tilde{d} = -\tilde{g}/\|\tilde{g}\|$ . One step of a shared length is then taken,

$$q \leftarrow q_0 + \alpha^* \tilde{d}. \quad (7)$$

The length  $\alpha^*$  comes from a line search that best decreases the exact CPML receiver objective along the exact CPML direction. The other methods take that same length along their own directions, and the new coefficients are scored with the exact objective. A step is harmful if that exact objective increases.

The full-state readout never raises the objective (0/100 harmful steps). Exact CPML is also 0/100, because the shared step length is the one that decreases the exact objective along the exact direction. The FNO candidate gradient raises the exact objective on 11/100 targets, and the receiver-only stencil raises it on 56/100. Recomputing the FNO Jacobian with a finite-difference increment of  $10^{-3}$  instead of the retained  $10^{-2}$  leaves the same 11/100 harmful targets, so the labels are not an artefact of that increment. The recount does not separate forward error from Jacobian error. Corollary 1 and Eq. (6) are why a forward-looking pressure match cannot stand in for that gradient.

The 11/100 and 56/100 failures are not a random scatter of bad steps. The stations sit in the top CPML layer, so a wrong memory history or a memory-free stencil can change the receiver derivative and send the exact objective up even when the pressure field looks accurate. For the FNO that happens mainly on layered models (7/25) and compact anomalies (4/25), with smooth and mixed classes at 0/25, while the receiver-only stencil is high in every class and worst on layered models

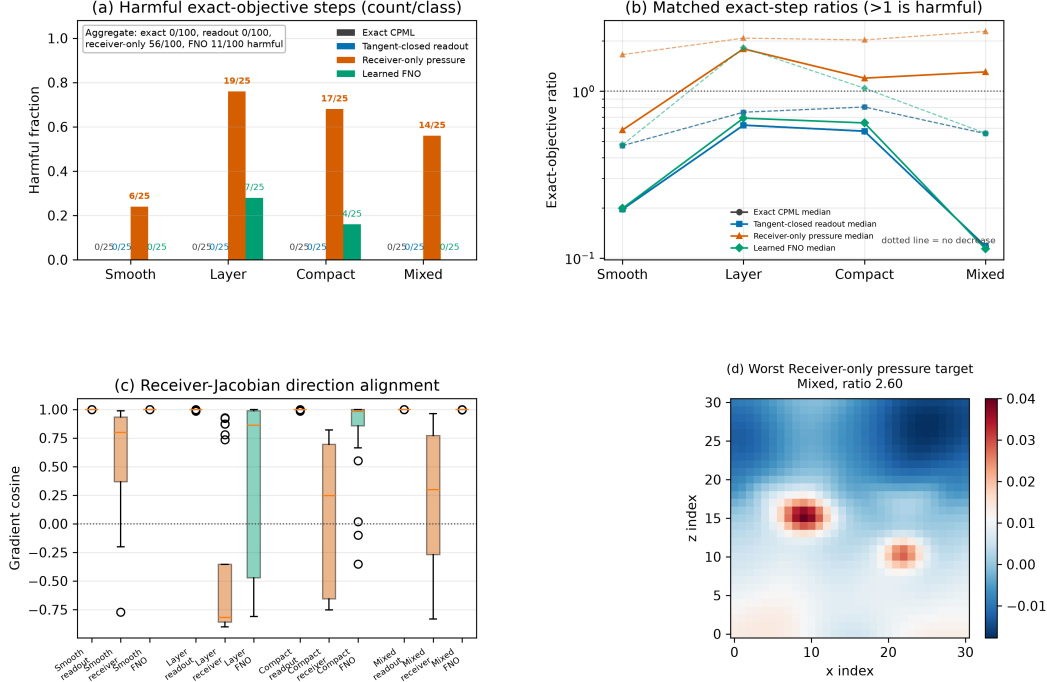


Figure 2: One matched FWI step on 100 targets with receivers in the CPML layer. Exact CPML takes 0/100 harmful steps by the shared step rule, the full-state readout takes 0/100, the FNO candidate gradient takes 11/100, and receiver-only pressure takes 56/100. The FNO count uses that arm’s forward map and Jacobian together.

(19/25). For that memory-free stencil, placing receivers at depths 2 and 3, still inside the top PML, gives 58/100 and 56/100 harmful steps, but moving them to depths 4 to 6 outside the layer drops the counts to 7/100, 3/100, and 5/100. The FNO arm is not part of that depth sweep, and target selection is reapplied at each depth. Noise replays and Wilson intervals are in Appendix C.

### 4.3 What evidence is enough for continuation?

The other use of a solver is to pick the run up in the middle, and that only works if the hidden edge memory travels with the pressure. Once pressure history, initial state, grid, and coefficients are fixed, the declared auxiliary updates determine matching memories causally. Replaying those updates through  $\mathcal{M}_h[p_\theta]$  turns pressure into a complete candidate state  $z_\theta = (p_\theta, \mathcal{M}_h[p_\theta])$  for the same state, residual, and tangent checks.

On the compact  $15 \times 15 \times 8$  test, a partitioned SIREN [Sitzmann et al., 2020] with this causal memory reconstruction passes the raw state screen across three seeds (state about 5.41–7.02%). Complete residuals 0.00825–0.01091 still fail  $10^{-4}$ , and active-family errors 11.6–15.1% fail the 10% sensitivity screen, so this is a limited state-metric result rather than a valid CPML continuation under the solver threshold. A separate  $31 \times 31 \times 8$  network with the same optimiser and width fails the raw state check even though pressure error stays near 1%. Architecture, parameter count, and absorber width also differ, so the pair is not a controlled grid-refinement result. We therefore keep compact neural CPML closure only as that small-grid state-screen result, with detailed errors in Appendix C.

The main continuation result uses a full-order Fourier/sine readout for the first slab. From that pressure and memory state, the solver advances seven more slabs using only the declared CPML recurrence. Across three seeds and 12 directions, carrying causal memory passes the forward, residual, and tangent checks through 64 updates, while resetting memory to zero fails them. A larger  $101 \times 101 \times 18$   $K=4$  experiment gives the same verdict across 36 source-seed-slab rows, so the later slabs continue when the absorber memories travel with the pressure.

Appendix C (Table 5) reports adaptive residual compression at retained spectral fractions from 75% to 95%, where forward complete-state checks still pass but most model tangents do not.

## 5 Discussion and limitations

The numerical tests confirm that a close pressure and receiver match can still fail the declared CPML recurrence, and that treating that match as a solver can change an FWI decision. The trained FNO looks accurate on snapshots and recorded data, but after we fill in its missing memories the complete stepper residual stays above both the loose diagnostic and the  $10^{-4}$  solver threshold (Table 1 and Figure 1). Using that candidate gradient for one matched FWI step raises the exact CPML objective on some of the 100 local targets whose receivers sit in the absorbing layer, a memory-free receiver stencil does so more often, and the full-state readout with the correct tangent does not (Figure 2). Exact CPML is also at 0/100 because the shared step decreases the exact objective. For the memory-free stencil, harmful steps are common when the stations sit inside the top PML and much rarer just outside it. The FNO was not included in that depth sweep.

This is expected once we remember what a receiver match actually shows. The predicted field, sampled at the same stations and processed the same way as the solver output, agrees with the recorded data. That is a check against the reference simulator, not proof that the surrogate is the simulator. Receivers are a thin slice of the grid, so they do not uniquely determine the hidden CPML memories that later updates depend on. Restart, replacement of the declared stepper, and the wavespeed derivative used by FWI therefore need the complete state and the tangent. Overclaiming is treating that match as if it were already enough to act on those stronger uses.

The other candidates make the same point in different ways. Adding the complete residual to the FNO pressure loss lowers the leftover residual but still not to  $10^{-4}$ . A POD–DeepONet-style pressure operator fails the same residual check, and a recurrent model that carries memory heads still does not close the recurrence or the tangent. On the continuation side, a small partitioned SIREN with memories reconstructed from pressure can pass the raw state screen on a  $15 \times 15$  grid while its complete residual still fails  $10^{-4}$ , and a separate  $31 \times 31$  network fails the raw state screen even though the pressure error stays near 1%. A full-order Fourier/sine readout continues through later slabs when absorber memories travel with the pressure, and fails when those memories are reset to zero.

On this local problem the Fortran CPML stencil is already cheap, so Figure 2 tests whether a candidate can stand in for it. A candidate still has to pass the  $10^{-4}$  residual and the wavespeed tangent before we treat it as that solver or as an FWI derivative.

The results are for acoustic CPML on local synthetic wavefields, with the grid, source, absorber, receivers, and thresholds fixed before scoring. The  $31 \times 31$  FNO and FWI cases are that local setting. The  $101 \times 101$  readout is continuation from a carried interface state, not a trained inversion at that scale. A pass here is not a field-scale FWI certificate (e.g. a survey with interior receivers, a different absorber thickness, or a production starting model). We also do not claim source-position transfer, and we do not claim that no architecture can close the declared recurrence, because those are separate questions from the local checks reported here.

## 6 Conclusion

Pressure is only part of the CPML state. We compared each candidate with the declared solver on wave kinematics, receiver data, pressure and memory, continuation, and the FWI derivative, and we reported the strongest use that evidence supported.

A trained FNO can look accurate on snapshots and receivers while still failing the complete recurrence after its missing memories are filled in, and its candidate gradient can raise the exact CPML objective on a local FWI step. A memory-free receiver stencil does so more often on the same targets. The full-state readout with the correct tangent does not, and exact CPML stays at zero because the shared step is chosen on the exact objective. A close pressure match does not show that the surrogate can restart the run or take the derivative the optimiser uses. Those uses need the hidden absorber state, the recurrence residual, and the wavespeed tangent.

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## A Proofs

**Lemma 1.** After spatial difference and absorber coefficient operators are frozen, the unsplit CPML memory update is affine in the current pressure and linear in the previous memory,  $\eta^{n+1} = B\eta^n + Cu^n$ . Unfolding  $q$  steps gives

$$\eta^n = B^q \eta^{n-q} + \sum_{r=0}^{q-1} B^r C u^{n-1-r}.$$

The pressure update is a function of  $(u^n, u^{n-1}, \eta^n)$  and the known source at time  $n$ . Hence  $x^n = (u^n, u^{n-1}, \eta^n)$  is Markov for the declared recurrence. A pressure-only pair  $(u^n, u^{n-1})$  omits  $\eta^n$ . If  $B$  is not nilpotent on the coupled subspace and  $C$  feeds back into the pressure stencil, there exist distinct memories  $\eta^n \neq \tilde{\eta}^n$  compatible with the same recent pressure window whose next pressure updates differ by  $B^q(\eta^{n-q} - \tilde{\eta}^{n-q})$ . Standard CPML memory factors  $e^{-(\alpha+\sigma/\kappa)\Delta t} \in (0, 1)$  are not nilpotent of any finite index. Only when that remainder is negligible can the pressure pair behave like an approximately Markov continuation state. On a damped  $31 \times 31 \times 18$ ,  $K=8$  replay, windows  $q = 1, 2, 4$  fail continuation (worst state/memory 31.7%, 21.3%, 12.9%).  $q = 8$  still has interface-memory error 15.6%. Passing  $q = 16$  on that replay is not a theorem that a fixed pressure archive is source-independent. Replaying  $\eta$  from  $t = 0$  using the entire pressure history recovers  $\eta^n$ , but the required archive length grows with  $n$  and is not a finite continuation state independent of the past source.

**Lemma 2** (Weaker screens don’t bound stronger errors). *Let the first arrival picker be unchanged by perturbations supported after an isolated arrival window, let  $P_R$  have a nontrivial kernel on the state space, and consider differentiable candidate maps without an a priori tangent stability bound. Then exact kinematic closure does not bound observation error. Exact observation closure does not bound state error. And exact state closure at one model  $m_0$  does not bound tangent error at  $m_0$ . Each uncontrolled error can be made arbitrarily large while the weaker error is zero.*

These are explicit existence constructions in an unrestricted candidate class, not a PDE certificate. For kinematics versus observations, let  $g$  be any nonzero receiver-data perturbation supported strictly after an isolated first-arrival picking window. Then  $\tilde{d} = d + ag$  has the same pick as  $d$  for every scalar  $a$ , while  $\|\tilde{d} - d\|_W = |a|\|g\|_W$  is unbounded. For observations versus state, choose nonzero  $v \in \ker P_R$  and set  $\tilde{u} = u + av$ . Receiver data are identical while state error grows as  $|a|$ . For state versus tangent, choose a bounded model-space functional  $\ell$ , a nonzero state  $h$ , and define  $\tilde{\mathcal{G}}(m) = \mathcal{G}(m) + a\ell(m - m_0)h$ . The maps agree at  $m_0$ , but their derivatives differ by  $a h \otimes \ell$ , whose operator norm is unbounded with  $|a|$ . A PDE constraint may shrink the admissible class. It does not restore an implication without a separate stability or observability theorem.

**Equation (4).** If the reference trajectory satisfies Eq. (3) and  $A_h$  is invertible,  $z_\theta - z_h = A_h^{-1}(A_h z_\theta - b_h)$ . Introducing the invertible state scaling  $S_Z$  and a left scaling  $D$  gives

$$S_Z(z_\theta - z_h) = (DA_h S_Z^{-1})^{-1} D(A_h z_\theta - b_h).$$

Taking Euclidean norms and using  $\|(DA_h S_Z^{-1})^{-1}\|_2 = \sigma_{\min}(DA_h S_Z^{-1})^{-1}$  yields Eq. (4). Here  $S_Z$  fixes the complete-state geometry and  $D$  is the inverse row- $\ell_2$  scaling of the assembled recurrence. Reported  $31 \times 31$  percentages use  $S_Z = I$ , so the bound is in the same packed recurrence coordinates

used by direct scoring. The residual includes source, initial, boundary, and memory rows because those rows define the declared CPML stepper. Residual and  $E_Z$  are scored separately, and the  $10^{-4}$  threshold is the solver threshold for this representation. The singular value below is a floating-point estimate for one retained  $31 \times 31$ , 18-update homogeneous calibration. It is not a certified lower bound, and that operator is not the 27-update heterogeneous FWI gate. Direct residual, state, tangent, and FWI checks remain the reported measurements.

The budget interpretation is as follows. With

$$\rho = \frac{\|D(A_h z_\theta - b_h)\|_2}{\|Db_h\|_2}, \quad C_Z = \sigma_{\min}(DA_h S_Z^{-1})^{-1} \frac{\|Db_h\|_2}{\|S_Z z_h\|_2 + \epsilon},$$

Eq. (4) gives  $E_Z \leq C_Z \rho$ . The factor  $C_Z$  includes the inverse scaled operator and the difference between residual normalisation and state-error normalisation, and it is tied to the declared  $S_Z$  and  $D$ . Thus  $\rho \leq 10^{-4}$  stays below the 10% reporting cut whenever  $C_Z \leq 10^3$  in that declared geometry.

The retained  $31 \times 31$ ,  $nt = 18$  homogeneous calibration recurrence with four PML cells is the operator used for this numerical estimate. Its row-scaled audit has state dimension 83,277, estimated  $\sigma_{\min}(DA_h) = 1.2736956443 \times 10^{-5}$ ,  $\|Db_h\|_2 = 0.8376993584$ , and  $\|z_h\|_2 = 497.3009028$  with  $S_Z = I$ . The resulting amplification factor is

$$C_Z = \frac{0.8376993584}{(1.2736956443 \times 10^{-5})(497.3009028)} \approx 132.25.$$

Consequently, that numerical estimate maps  $\rho = 10^{-8}$  to  $E_Z \leq 1.32 \times 10^{-6}$ ,  $\rho = 10^{-4}$  to  $E_Z \leq 1.32 \times 10^{-2}$ ,  $\rho = 10^{-3}$  to  $E_Z \leq 1.32 \times 10^{-1}$ , and  $\rho = 10^{-2}$  to  $E_Z \leq 1.32$ . These maps calibrate the retained homogeneous recurrence. They do not certify the FWI gate. We treat  $10^{-8}$  as a numerical-sanity level,  $10^{-4}$  as the solver threshold for the declared representation, and  $10^{-3}$  and  $10^{-2}$  as sensitivity and diagnostic levels.

**Corollary 1.** Differentiate Eq. (3) in direction  $v$  to obtain  $A_h \dot{z}_h + \dot{A}_h z_h = \dot{b}_h$ . Subtract the differentiated residual definition  $r_{J,h} = A_h \dot{z}_\theta + \dot{A}_h z_\theta - \dot{b}_h$  and rearrange.

$$A_h(\dot{z}_\theta - \dot{z}_h) = r_{J,h} - \dot{A}_h(z_\theta - z_h).$$

The same scaling argument as for Eq. (4) produces Eq. (5). If  $r_{J,h} = 0$  and  $z_\theta = z_h$ , then  $\dot{z}_\theta = \dot{z}_h$ . Vanishing  $A_h z_\theta - b_h$  alone does not force  $r_{J,h} = 0$ .

**Equation (6).** Let  $\Phi(m) = \frac{1}{2}\|F(m) - d\|_W^2$  and  $\tilde{\Phi}(m) = \frac{1}{2}\|\tilde{F}(m) - d\|_W^2$  with  $F = P_R \mathcal{G}_h^{\text{CPML}}$  and  $\tilde{F}$  the candidate reduced map. Then  $\nabla \Phi = J^T W(F - d)$  and  $\nabla \tilde{\Phi} = \tilde{J}^T W(\tilde{F} - d)$ . Substitute  $\tilde{F} = F + (\tilde{F} - F)$  and expand.

**Fixed map nonequivalence of generic absorbers and CPML.** For a fixed model and discretisation, both initialised linear recurrences define source-to-pressure maps. If they agreed for every source in a linear source family, their frequency-domain transfer operators would agree on every frequency excited by that family. CPML replaces boundary-normal derivatives by rational factors containing  $(\alpha + i\omega)^{-1}$ . Inverse transformation introduces the auxiliary memory state. A scalar sponge has polynomial temporal dependence, and a sampled local radiation penalty introduces no matching memory recurrence. Under a spectral-separation assumption on an excited frequency interval, the transfer operators differ, so at least one source produces different pressure data. Interior residual minimisation alone leaves the boundary data undetermined.

## B Experimental protocol

**Design principles.** Each check is scored separately against the declared SEISMIC\_CPML setup. Kinematics, observation, complete state, continuation, and FWI derivative are not inferred from one another. A residual is the residual of the complete recurrence, including memory rows. The solver threshold used throughout is  $10^{-4}$ . Table 2 also reports  $10^{-2}$  as a  $100\times$  looser diagnostic column. Figure 2 labels a step harmful when the exact objective increases.

**Profile sanity check.** Before scoring learned surrogates, we run six deterministic candidates with known outcomes. The cases cover correct CPML, exact pressure with fields withheld, exact pressure with zeroed memories, correct terminal state with wrong continuation, correct forward fields with wrong tangent, and a memory-free recurrence scored under the CPML recurrence. The profile returns the expected strongest use on every row, with 0/6 cases where a weaker test was accepted as a stronger use and 0/6 cases where a valid stronger use was rejected. Six of six deliberately withheld optional fields are reported as unmeasured rather than inferred from weaker evidence.

**Intended use.** The experiments score a physics-informed neural network (PINN) or operator candidate against the declared SEISMIC\_CPML setup, covering the source family, acquisition, horizon, and norm. The reported conclusion names the strongest use that passed and withholds stronger uses. A local  $31 \times 31$  pass should not be read as production FWI.

**Performance on this release.** The FNO residual 0.0283 fails  $10^{-2}$  after causal completion, where causal completion means reconstructing  $\mathcal{M}_h[p]$  from predicted pressure by replaying the declared memory recurrence (memory reconstruction below). The residual-trained control reaches 0.00672 and still fails  $10^{-4}$ . Figure 2 reports exact CPML 0/100, which follows because the shared step length decreases the exact objective along the exact direction, full-state readout with the correct tangent 0/100, FNO candidate gradient after causal completion 11/100, and receiver-only memory-free stencil 56/100. The 100 targets are those kept by the frozen readout-error floor below. Wilson 95% intervals are 0 to 3.7% for each zero-harm arm, 6.3 to 18.6% for the FNO arm, and 46.2 to 65.3% for the receiver-only arm. These are nominal binomial intervals on this selected sample, not calibrated population rates, and the exact-CPML zero is structural under the step rule. A paired bootstrap over matched target rows gives receiver-only minus FNO excess harm 0.45 (95% interval 0.35 to 0.55). Table 3 gives the gradient-error recount versus exact  $\nabla\Phi$ .

Table 2 is a threshold-sensitivity check from retained scalar measurements. It reads as separation evidence for the declared representation and does not rerun CPML, FWI, or FNO training. The retained  $31 \times 31$  homogeneous calibration above explains the scale because the numerical estimate puts  $10^{-4}$  below the 10% reporting screen on that recurrence, while  $10^{-3}$  is already above that screen under the same estimate. The learned pressure-surrogate rows remain above  $10^{-3}$ . The residual-trained FNO passes the loose  $10^{-2}$  diagnostic but fails  $10^{-3}$  and the declared  $10^{-4}$  threshold. Positive-control residuals are near  $10^{-10}$  and pass even the stricter  $10^{-8}$  sanity level. The  $10^{-4}$  column therefore sits in the observed gap between learned surrogates and attainable controls. The  $10^{-2}$  column is only a progress diagnostic that asks whether residual training has reached a coarse one-percent complete-recurrence regime, not whether the candidate has earned solver status.

Table 2: Residual-threshold sensitivity from retained scalar measurements. P/F denotes pass/fail at that threshold. The  $10^{-4}$  column is the learned-solver CPML threshold for the declared representation. The  $10^{-3}$  column is the first listed relaxation whose numerical  $C_Z$  estimate exceeds the 10% reporting screen, and  $10^{-2}$  is a loose diagnostic column.

| Candidate          | Residual                | $10^{-8}$ | $10^{-4}$ | $10^{-3}$ | $10^{-2}$ | $10^{-1}$ |
|--------------------|-------------------------|-----------|-----------|-----------|-----------|-----------|
| FNO MSE            | $2.831 \times 10^{-2}$  | F         | F         | F         | F         | P         |
| FNO residual 1     | $6.721 \times 10^{-3}$  | F         | F         | F         | P         | P         |
| FNO residual 2     | $6.610 \times 10^{-3}$  | F         | F         | F         | P         | P         |
| POD DeepONet worst | $1.298 \times 10^{-2}$  | F         | F         | F         | F         | P         |
| FNO corrected      | $6.348 \times 10^{-10}$ | P         | P         | P         | P         | P         |
| Full readout       | $3.976 \times 10^{-10}$ | P         | P         | P         | P         | P         |

**Experimental records.** The benchmark cases, profile sanity check, and scoring records define the local replay setup used for the reported tables and figures. The public version will include machine-readable records and scripts for the lightweight checks. Heavyweight neural checkpoints and state/Jacobian archives are separate optional reproducibility inputs.

**Reference solver.** All primary experiments use the unsplit CPML pressure formulation of Komatitsch and Martin [2007] as implemented in SEISMIC\_CPML [Komatitsch and Martin, 2018], with declared grid, Courant number, PML thickness, source wavelet, and receiver processing frozen before scoring. Candidates are never scored against a different absorber than the one declared for the experiment.

**24D wavespeed parameterisation.** The local FNO and FWI experiments use a fixed 24-coefficient multiplicative speed family. We start from the smooth-layered background speed  $c_0(x, z)$  and build a dictionary of spatial perturbation fields on the same grid, including a constant field, smooth sinusoidal modes, layered step and band fields at four depths, compact Gaussian anomalies, and seeded smooth random cosine fields. Each raw field is normalised by its maximum absolute value. The first 24 dictionary fields are vectorised and QR-orthonormalised, giving basis columns  $\{b_k\}_{k=1}^{24}$ . A coefficient vector  $q \in \mathbb{R}^{24}$  defines

$$c_q(x, z) = c_0(x, z) \left[ 1 + \delta \frac{\sum_{k=1}^{24} q_k b_k(x, z)}{\left\| \sum_{k=1}^{24} q_k b_k \right\|_{\infty}} \right],$$

with  $\delta$  equal to the declared relative velocity amplitude. Training and held-out draws sample  $q$  from the seeded standard normal generator. Target selection for the FWI matched-step experiment is then applied with the frozen readout-error floor described below.

**Trained FNO overclaiming case.** FNO2d with time-channel output has 1,188,156 parameters, four layers,  $12 \times 12$  modes, width 32, 2000 AdamW epochs with cosine decay, learning rate 0.002, weight decay  $10^{-6}$ , and mini-batch 32. Checkpoint selection uses the lowest interior snapshot error on a held-out 24D val speed (seed 20260817) unused in the reported test rows (best epoch 1875). The training set contains 387 CPML pressure fields on the FWI experiment setup ( $31 \times 31$ , 27 updates, frequency 10, source abscissae  $\{0.25, 0.50, 0.75\}$ ), using the unperturbed smooth-layered background and 128 random 24D coefficient draws in the same QR basis. The network uses declared zero initial pressure and an exterior-boundary pressure mask applied after the FNO output. No CPML memory labels are used. The held-out rows are two unused 24D speed draws at the centre shot (seeds 20260815 and 20260816). The main FNO experiment scores memory reconstruction of the first of those pressures by replaying  $\mathcal{M}_h[p]$  from the predicted pressure field. The zero-memory lift is a control series in the retained figure. Figure 2’s learned-FNO arm uses the central-difference Jacobian of this checkpoint through the same causal packing (retained increment  $10^{-2}$ , checked at  $10^{-3}$ ). The records also include a residual-corrected completion of the same pressure (relative residual  $6.35 \times 10^{-10}$ ). That post-hoc projection is a different claim because it projects onto the declared recurrence rather than testing a learned solver, and it is not used in the main FNO experiment or Figure 2.

**Residual-trained FNO control.** Section 4 reports both unused centre-shot draws. Same FNO2d, 387 fields, 2000 AdamW epochs, and val-snapshot rule, retrained with pressure MSE and the scaled complete CPML residual of  $z = (p, \mathcal{M}_h[p])$  (weight 1). Memories remain causal functionals of predicted pressure. No CPML memory labels are used. Independent memory-output heads are withheld because they can break the memory recurrence. Checkpoint selection remains lowest interior snapshot error, as for the pressure-MSE FNO. Residual selection on the logged val residual (epoch 1 and every 25 epochs, 81 evals) picks the same epoch 2000 (val residual  $5.86 \times 10^{-3}$ ), so the residual-trained rows are that iterate. After causal completion, memory-update residual rows are  $\sim 10^{-15}$ . The leftover complete residual is the pressure stencil. Draw 2 residual-trained tangent is 7.61%. This checkpoint is not substituted into the main FNO experiment or Figure 2, and its Jacobian is not recomputed.

**Constructed pressure-only control.** Grid  $31 \times 31$ , 27 stored time levels, homogeneous acoustic medium, and one shallow source near the top absorber. The candidate is a memory-free pressure stencil with no CPML auxiliaries in the update, marched exactly, then followed by memory reconstruction from that pressure. It is not CPML pressure with rebuilt memories. Neural optimisation is removed. This is the receiver-only FWI arm, not the FNO residual gap. Ordinary checks include physics residual, pressure snapshots, identically processed source-depth gather, and a restricted receiver-tangent probe. Complete-state and complete-state-tangent checks use the concatenated pressure and memory metric of Eq. (2). The operational reporting cut is 10% (justified below). The same candidate is also scored at 1%, 5%, and 20% in the experimental records.

**FWI matched-step experiment.** The test is a 24-coefficient local matched-step experiment on a smooth-layered heterogeneous CPML medium on  $31 \times 31$  with 27 updates and four PML cells (interior  $23 \times 23$ ). Acquisition uses three shallow sources, nine sparse near-surface receivers per shot

at depth index 3 (inside the top PML), alternate-time samples, and additive Gaussian receiver noise at 5% of the clean target-minus-background receiver difference, not 5% of the full gather, unless swept.

The cutoff  $2.462 \times 10^{-4}$  equals the retained full-order readout receiver error and was frozen before the reported scoring. A draw is kept only if its clean exact-reference receiver residual norm exceeds that cutoff. It is not an independent instrument floor. That rule drops 45 weak-signal draws (27 layered, 18 compact) before filling 25 in each of smooth, layered, compact-anomaly, and mixed classes. Each candidate gradient  $\tilde{g}$  on the coefficient vector  $q$  uses the unit-direction update of Eq. (7). The length  $\alpha^*$  is the best exact-CPML line-search step along the exact direction, using 45 logarithmically spaced trial lengths from  $10^{-4}$  to 1.5 times the target coefficient norm. Every other arm takes that same length along its own unit direction, so exact CPML cannot take a harmful step under this rule. The new coefficients are scored with the exact CPML receiver objective. A step is harmful if that objective increases. The four arms are exact CPML, a full-state readout with the correct tangent, receiver-only pressure (memory-free stencil, then causal  $\mathcal{M}_h$ ), and the matched FNO arm (finite-difference Jacobian together with the FNO forward map, causal packing). The retained FNO Jacobian uses a central-difference increment of  $10^{-2}$ . Repeating that Jacobian at an increment of  $10^{-3}$  changes it by  $1.32 \times 10^{-3}$  relative Frobenius with zero harmful-label changes, while exact CPML Jacobians at those increments differ by  $1.23 \times 10^{-6}$ . That check does not separate forward bias from Jacobian error.

The five 5% noise-seed replays reuse the retained targets, so they test receiver-noise stability rather than independent target prevalence. A 0%, 5%, and 10% noise-fraction sweep is reported separately. Copying those cell counts onto  $51 \times 51$  thins the absorber and moves the same physical receivers out of the PML, where receiver-only harm is then 0/100. That 0/100 is the interior-receiver case on this stencil, not a larger-grid cure. Matching physical PML thickness and horizon (45 updates, 7 PML cells) puts receivers back in the layer, where receiver-only harm is 56/100, the full-state readout is omitted, and ordinary receiver error is 13.3%. The  $31 \times 31$  receiver-depth sweep is the memory-free stencil, not the FNO. It gives 58/100 and 56/100 receiver-only harm at depth indices 2 and 3 inside the top PML, but only 7/100, 3/100, and 5/100 at depths 4 to 6 outside it. Depth 1 cannot fill the fixed cutoff quota, and selection is reapplied at each depth. A  $51 \times 51$  factor sweep gives 11/100 harm for matched horizon with thin PML and 62/100 for matched PML thickness with short horizon. Neither recipe is a field-scale FWI survey.

**Reporting cut and gradient-error recount.** The 10% cut is a named reporting threshold for relative  $L^2$  state, memory, and state-tangent checks, while Figure 2 harm counts use exact-objective increase. Section 4 already names the 4.79% FNO tangent as the reason a 1% state/tangent cut would hide the misleading case. We report 10% because ordinary relative-error tables would still accept the trained FNO after memory reconstruction (state 0.253%, memory 0.253%, state tangent 4.79%), while the constructed memory-free stencil fails it (state 29.8%, tangent 33.7%). A 5% state/tangent cut would still accept the completed FNO. A 1% cut would reject it before the residual and FWI distinction is visible. The retained numerical  $C_Z$  estimate is consistent with this separation on the 18-update homogeneous calibration, because  $10^{-4}$  is below the 10% state screen in that geometry, while  $10^{-3}$  is already above it under the same estimate. It is not a certified bound for the FWI operator.

Table 3 and Figure 3 recount each candidate FWI gradient against the exact CPML  $\nabla\Phi$  at  $\tau \in \{1, 5, 10, 20\}\%$ . That comparison is not  $E_{J,Z}$ . Harmful steps among targets below the cut remain 0 at all four  $\tau$ . Receiver-only is above the cut on 100% (including 56/100 harmful rows), and the FNO is above the cut on 38% at 10% (including all 11 harmful steps). At 1% the FNO is above the cut on 100% and the full-state readout loses 21 targets, so that cut mixes gradient noise with map error. At 20% the FNO is below the cut on 67% versus 62% at 10%, still with 0 remaining harm.

A controlled prescribed error sweep has 0% harmful matched steps at 1, 5, 10, and 20%, and first records harmful steps at 50% (1.5% of 200 targets). Residual scoring remains  $10^{-4}$  even when a target sits below the  $\nabla\Phi$  recount cut.

**Compact neural slab.** Pressure is a partition-of-unity SIREN [Sitzmann et al., 2020]. Memories are exact causal functionals of that pressure (no CPML labels). The optimiser uses 1,000 Adam steps and then 5,000 L-BFGS steps, partition width 33, and three seeds {8210, 8211, 8212}. The

Table 3: Gradient-error recount of the Figure 2 matched-step experiment versus exact CPML  $\nabla\Phi$ . Cells are targets whose candidate-gradient relative error is at most  $\tau$  (harmful among those). Unfiltered exact-objective harmful counts stay 0/100, 0/100, 56/100, and 11/100.

| Method                 | 1%      | 5%      | 10%     | 20%     | Unfiltered harm |
|------------------------|---------|---------|---------|---------|-----------------|
| Exact CPML             | 100 (0) | 100 (0) | 100 (0) | 100 (0) | 0%              |
| Full-state readout     | 79 (0)  | 90 (0)  | 92 (0)  | 98 (0)  | 0%              |
| Receiver-only pressure | 0 (0)   | 0 (0)   | 0 (0)   | 0 (0)   | 56/100          |
| Learned FNO            | 0 (0)   | 53 (0)  | 62 (0)  | 67 (0)  | 11/100          |

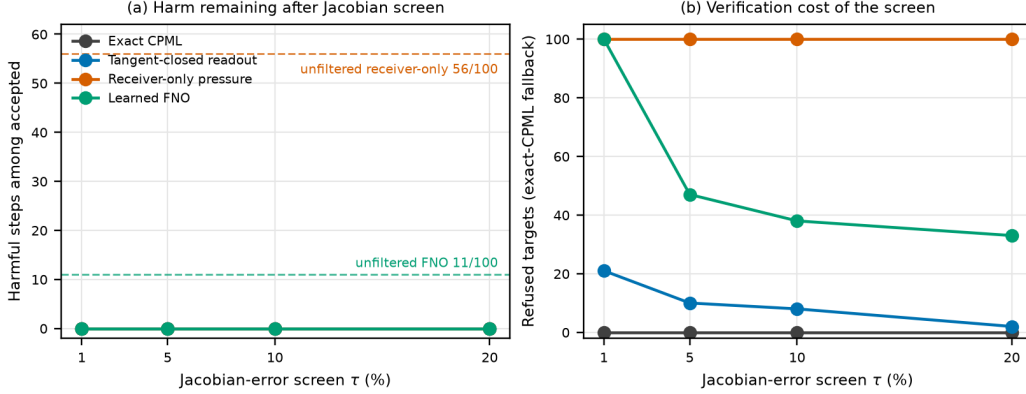


Figure 3: Gradient-error recount versus exact CPML  $\nabla\Phi$  on the 100 retained targets. Solid markers are harmful steps among targets with gradient relative error at most  $\tau$ . Dashed lines are the Figure 2 counts. Panel (b) is the number of targets above the cut.

$15 \times 15 \times 8$  case passes the raw state screen and fails the  $10^{-4}$  residual. The  $31 \times 31 \times 8$  case fails the raw state screen. These are not a matched grid-refinement pair.

**Full-order readout.** Lift-free Fourier/sine readout on a  $31 \times 31 \times 8$  slab with  $K=8$  retained spectral modes, solved by LSQR [Paige and Saunders, 1982], then seven additional slabs advanced only by the declared recurrence from the carried interface state (pressure and causal memories). Scoring uses three seeds and 12 model directions, with forward, residual, and tangent checks. Scale support uses  $101 \times 101 \times 18$ ,  $K=4$ , three sources, three seeds, four slabs, and eight directions. The reported 36-row pass is  $3 \times 3 \times 4$  source-seed-slab rows. Eight directions are scored inside each row, not as extra rows. The compression frontier is adaptive residual truncation of the  $31 \times 31 \times 18$  readout at retained spectral fractions 75, 85, 90, 95, 100%, with the same 12 tangent directions.

**Compute.** Retained timings use an Apple M1 Max CPU. Direct CPML marches on the common 18-update slab take median  $8.2 \times 10^{-4}$  s on  $31 \times 31$  (seven repeats). Full-order LSQR readout on the same slab takes median 10.4 s. The same-hardware stacked CPML tangent is 7.54 s. Neural cold-start jobs are single-CPU training runs on the order of minutes ( $15 \times 15$ ) to tens of minutes ( $31 \times 31$ ). The 100-target FWI experiment and  $51 \times 51$  absorber-sampling control were run on CPU cluster nodes (no GPU). Table 4 copies the lift-free Fourier/sine LSQR versus march scaling on that 18-update slab. The  $31^2$  LSQR wall 10.6 s is the worst seed row.

Table 4: LSQR readout check versus CPML march on the common 18-update slab. LSQR wall is the worst row across seeds 8210 to 8212. March time is the median of seven reference marches on the same CPU. The readout-check to march ratio grows with grid size.

| Grid             | LSQR wall (s) | CPML march (s)        | LSQR / march      |
|------------------|---------------|-----------------------|-------------------|
| $31 \times 31$   | 10.6          | $8.17 \times 10^{-4}$ | $1.3 \times 10^4$ |
| $51 \times 51$   | 21.7          | $1.39 \times 10^{-3}$ | $1.6 \times 10^4$ |
| $101 \times 101$ | 75.3          | $4.21 \times 10^{-3}$ | $1.8 \times 10^4$ |

## C Additional results

**Compact SIREN  $31 \times 31$  rescue.** Zero-start compact SIRENs on the lift-free  $31 \times 31 \times 8$  causal-memory setup fail the complete-state check across three seeds. Fitting a compact Fourier-feature SIREN to a lift-free spectral pressure readout, followed by absorber-weighted polish, puts pressure at 0.0227 to 0.0360 and receiver at 0.00640 to 0.0142. Memory stays 11.9 to 15.4, and raw state stays 2.87 to 3.70. The compact bottleneck is not the causal memory formula. Pressure must control PML-band derivatives tightly enough for  $\mathcal{M}_h[p]$ .

**Noise seed and class-wise FWI structure.** At 5% noise, harmful counts for receiver-only pressure across five seeds are 56, 56, 55, 56, 55 out of 100 (median 56/100). The matched FNO arm has counts 11, 11, 11, 11, 11 out of 100 (median 11/100). The harmful FNO target set is the same 11 rows on every seed, with seven layered and four compact cases. These seed rows reuse the same retained target list and are not pooled as independent target sets. Exact CPML and the full-state readout with the correct tangent remain 0/100 on every seed.

The 0%, 5%, and 10% noise-fraction sweep gives 55/100, 56/100, and 56/100 for receiver-only pressure and 11/100, 11/100, and 10/100 for the FNO. Among the 11 retained-seed FNO-harmful targets the exact-objective ratios have median 1.48, 90th percentile 1.87, and max 1.97 (nine exceed 1.2, and nine have negative gradient cosine). The all-target 90th percentile of 1.007 is the start of that 11-target harmful tail, not a summary of the 89 non-harmful cases. One compact ratio is 1.0004. The other ten are not threshold noise.

Class-wise harmful counts at the retained 5% seed are receiver-only smooth 6/25, layered 19/25, compact 17/25, mixed 14/25, and learned FNO smooth 0/25, layered 7/25, compact 4/25, mixed 0/25 (11/100 aggregate). Harmful matched steps concentrate in bins whose observed receiver-Jacobian error versus exact  $\nabla\Phi$  exceeds 10%.

**Nonlinear compact-anomaly loop.** A short nonlinear loop on a compact low-velocity anomaly, using the same  $31 \times 31$  acquisition as the local experiment, recovers the anomaly under exact CPML and under the full-state readout with the correct tangent. Receiver-only pressure stalls despite initial receiver error below 10%. This is a workflow illustration of Figure 2, not an application-scale inversion.

**Compression tangent frontier.** Table 5 reports the  $31 \times 31 \times 18$  adaptive-residual readout. Forward complete-state checks stay closed from 75% retained spectral fraction, but directional tangent checks recover only at full order.

Table 5: Compression and tangent frontier on the  $31 \times 31 \times 18$  adaptive-residual readout. Forward complete-state pass rate stays 1.0 until the last row, but tangent closure appears only at full order. The last column is relative  $L^2$  (operational cut 0.10). The 95% row’s worst-direction error 8.98 is therefore a fail, which is why the pass count stays 33%.

| Retained % | Params | Forward pass | Tangent pass | Max rel. state tangent |
|------------|--------|--------------|--------------|------------------------|
| 75         | 11372  | 1.00         | 33%          | 21.6                   |
| 85         | 12886  | 1.00         | 33%          | 38.6                   |
| 90         | 13643  | 1.00         | 33%          | 22.5                   |
| 95         | 14400  | 1.00         | 33%          | 8.98                   |
| 100        | 15156  | 1.00         | 100%         | $2.74 \times 10^{-3}$  |

**Constructed pressure-only mechanism.** Figure 4 shows the memory-free stencil used as the receiver-only FWI arm in Figure 2 (56/100 harmful steps). After memory reconstruction, the FNO passes the 10% relative-error cuts and is rejected by residual 0.0283 failing  $10^{-2}$  (11/100 harmful steps).

**POD–DeepONet corroboration.** An independent POD–DeepONet-style pressure operator (165,763 parameters, POD rank 8) is fit by ridge regression on the same centre-source heterogeneous-lens family. Across three held-out rows, ordinary pressure/receiver/snapshot checks

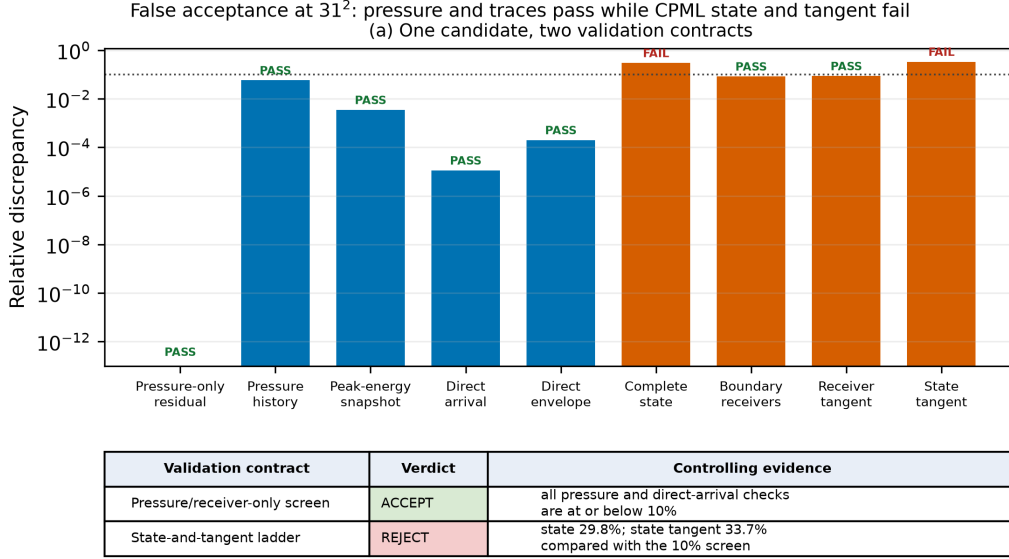


Figure 4: Constructed pressure-only control. The candidate is a memory-free pressure stencil marched exactly, not a trained network and not CPML pressure. Ordinary residual, pressure, receiver, and restricted receiver-tangent checks pass, but complete CPML state and state tangent fail the 10% cut (state  $\approx 29.8\%$ , state tangent  $\approx 33.7\%$ ). Omitting absorber memories is sufficient for that gap, which is Figure 2’s receiver-only arm.

pass all three (worst 1.57%/1.08%/1.58%). Memory reconstruction still fails  $10^{-4}$  on all three (worst relative residual 0.0130) and fails the 10% state-tangent cut on one of three (worst 12.2%). Zero-memory lift remains a  $\sim 100\%$  control. The purpose is to test ordinary checks on pressure operators, not to rank architectures.

**State-aware recurrent operator.** The explicit state-aware baseline has pressure,  $\psi_x$ ,  $\psi_z$ ,  $\chi_x$ , and  $\chi_z$  output heads and carries its predicted memories through a free rollout. It has 1,186,629 parameters and is trained on one-step complete-state targets and a six-step rollout term. Across three held-out 31<sup>2</sup>, 27-update rows, the free rollout has pressure error 0.08196 to 0.2186, state error 0.0500 to 0.1507, memory error 0.0500 to 0.1507, complete residual 8.81 to 9.31, and state-tangent error 0.160 to 0.346. The zero-memory restart ablation keeps the same pressure but has memory error 1.00 and state error about 1.00. Thus explicit memory carry helps the hidden-state metric, but it still does not pass the declared CPML recurrence or tangent checks.

**Held-out source-position FNO.** The main FNO experiment scores an unused 24D speed draw at a training source abscissa. We don’t claim source-position transfer.

**Held-out sources (readout family).** Eight held-out source positions times three seeds, and a heterogeneous five-source tangent-stress experiment, preserve the causal-memory versus zero-memory ordering outside the showcased homogeneous source set. These retained checks support the ordering reported here.

**Reproducibility.** We retain machine-readable benchmark rows, scoring records, scripts, figure sources, manifest hashes, and per-record reproduction status for the public version. These records cover the benchmark cases, residual-threshold sensitivity, FWI summary, profile sanity check, sigma-min and  $C_Z$  scalar audits, FNO checkpoints, state/Jacobian caches, and readout solve archives used to reproduce the reported numbers. From-scratch FNO retraining is treated as a robustness path rather than exact paper-number replay because neural training can drift across hardware and PyTorch builds. All reported wavefields are synthetic. SEISMIC\_CPML is used under its upstream GPL terms and cited as both the CPML method paper [Komatitsch and Martin, 2007] and the pinned software source [Komatitsch and Martin, 2018].